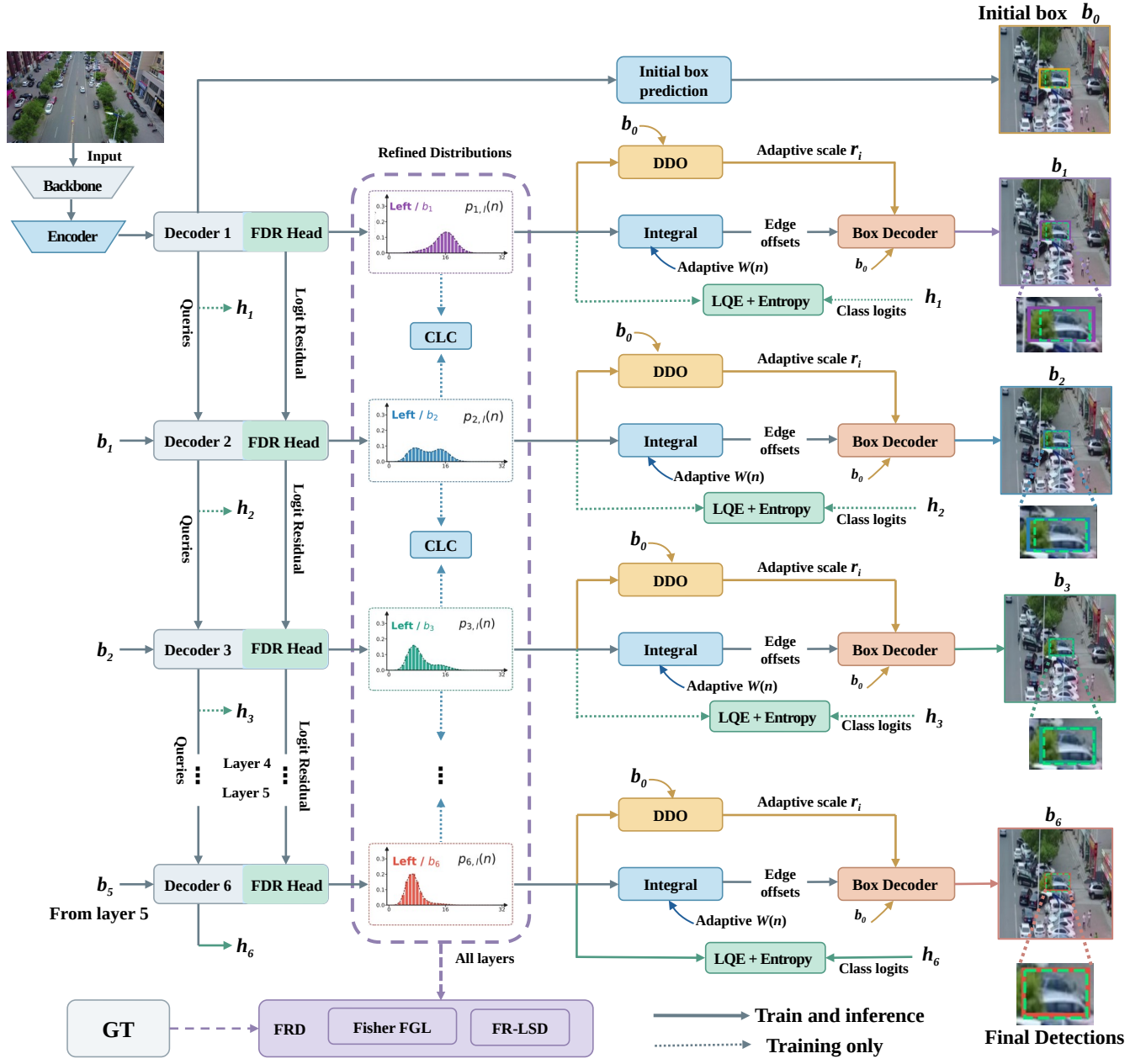
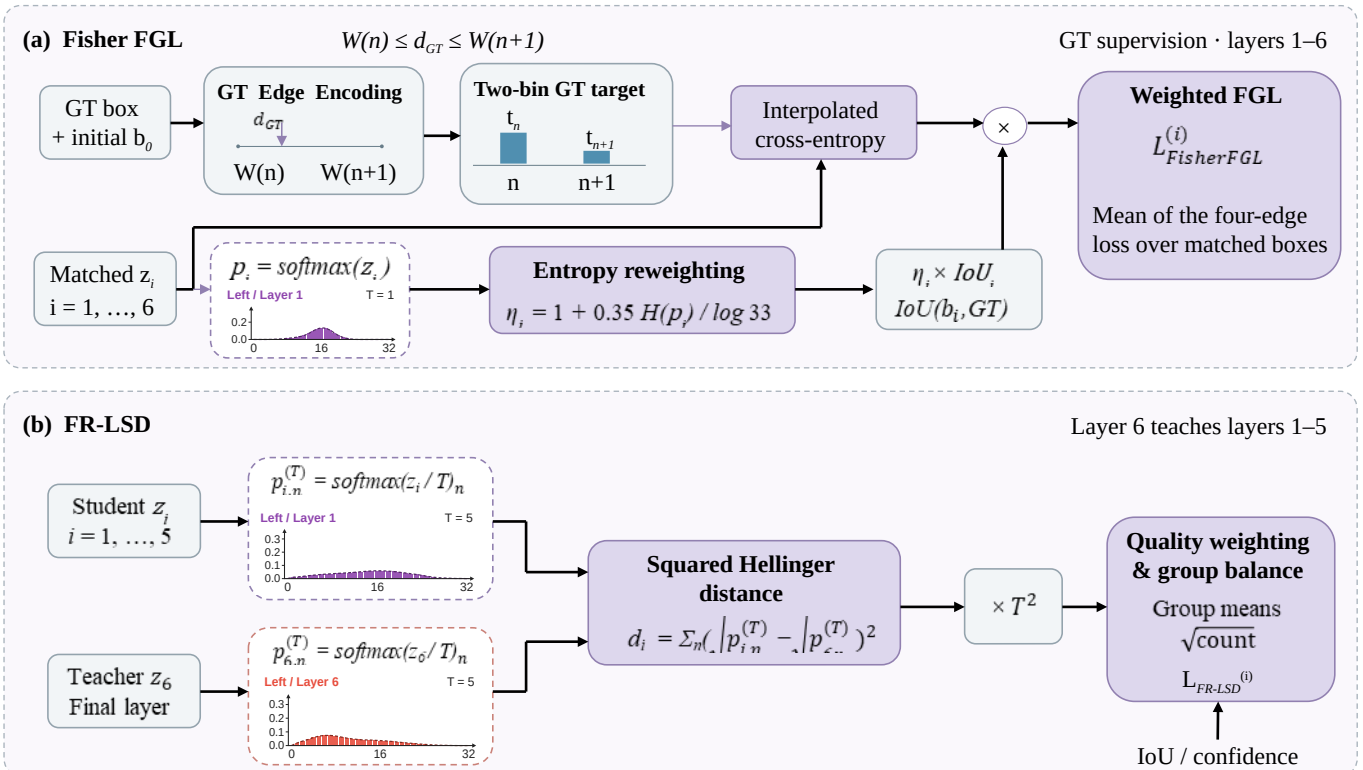
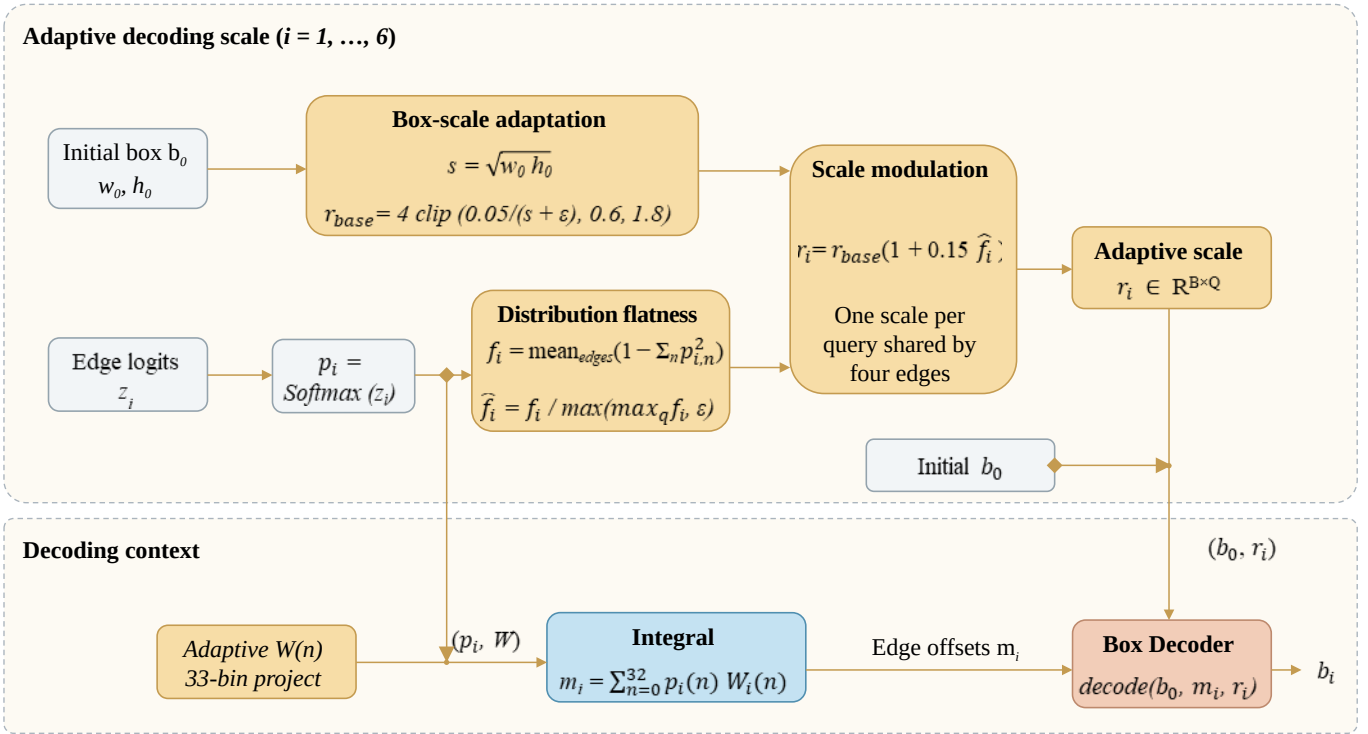

Initial box b_0
Input
Backbone
Encoder
Initial box prediction
Refined Distributions
Decoder 1
FDR Head
Queries
h_1
Logit Residual
b_1
Decoder 2
FDR Head
Queries
h_2
Logit Residual
b_2
Decoder 3
FDR Head
Queries
h_3
Layer 4
Layer 5
Logit Residual
b_5
From layer 5
Decoder 6
FDR Head
h_6
Left / b_1 $p_{1,i}(n)$
Left / b_2 $p_{2,i}(n)$
Left / b_3 $p_{3,i}(n)$
Left / b_6 $p_{6,i}(n)$
CLC
CLC
All layers
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_1
b_1
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_2
b_2
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_3
b_3
b_0
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
b_0
LQE + Entropy
Class logits
h_6
b_6
Final Detections
GT
FRD
Fisher FGL
FR-LSD
Train and inference
Training only

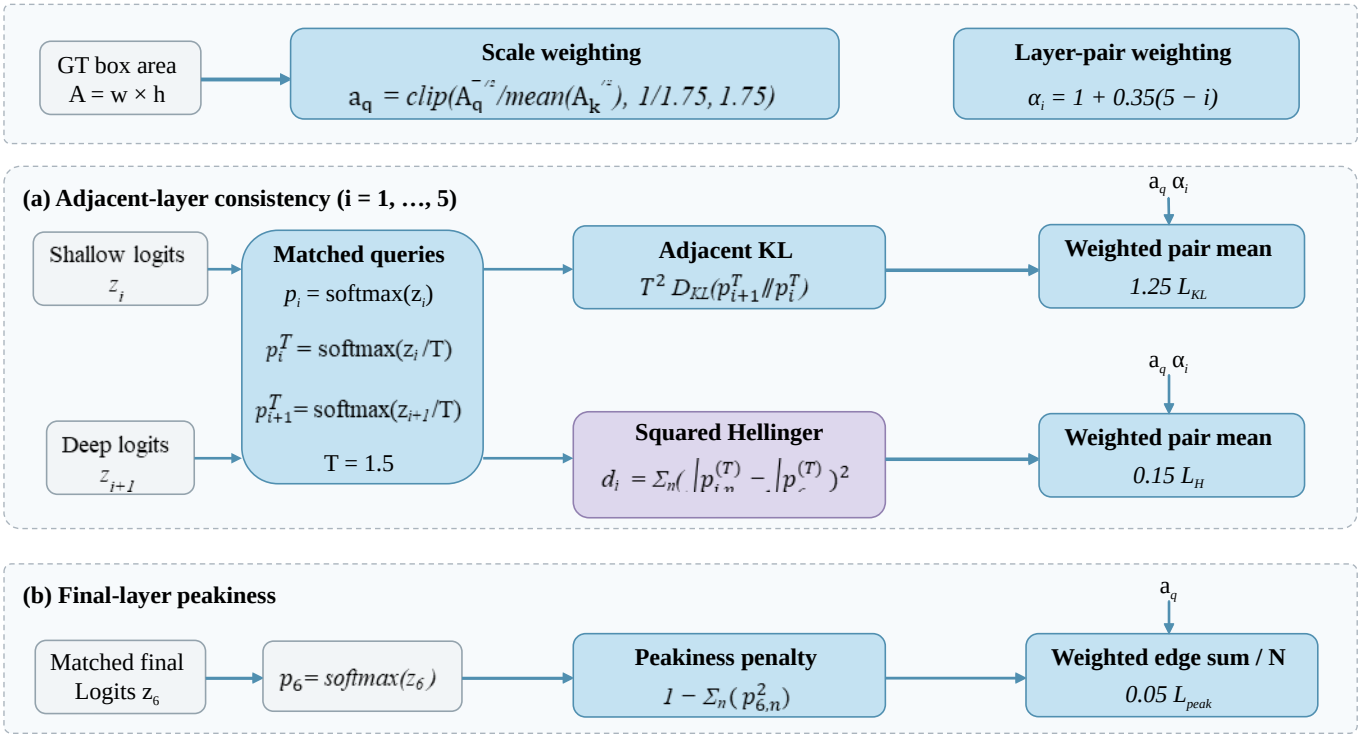
Fisher–Rao Distribution Loss



Decoder Distribution Optimizer (DDO)



Cross-Layer Consistency



Localization Quality Estimator (LQE)

